

UNIT-V: Numerical methods-1

Newton's interpolation formulae

1. The table gives the distances in nautical miles of the visible horizon for the given heights in feet above the earth's surface:

X=height	100	150	200	250	300	350	400
Y=distance	10.63	13.03	15.04	16.81	18.42	19.90	21.27

Find the values of y when (i) $x = 160$ ft, (ii) 410 ft.

2. From the following table, estimate the number of students who obtained the marks between 40 and 45:

Marks:	30-40	40-50	50-60	60-70	70-80
No. Of students	31	42	51	35	31

3. Find the cubic polynomial which takes the following values:

x:	0	1	2	3
f(x):	1	2	1	10

Hence find $f(4)$.

4. In the table below, the values of y are consecutive terms of a series of which 23.6 is the 6th term. Find the first and tenth terms of the series:

x:	3	4	5	6	7	8	9
y:	4.8	8.4	14.5	23.6	36.2	52.8	73.9

5. Calculate the value of $f(1.85)$, given the following table:

x:	1.7	1.8	1.9	2.0	2.1	2.2	2.3
f(x):	5.474	6.050	6.686	7.389	8.166	9.025	9.974

6. Express the value of θ in terms of x using the following data:

x:	40	50	60	70	80	90
θ :	184	204	226	250	276	304

Also find θ at $x = 43$.

7. Given $\sin 45^\circ = 0.7071$, $\sin 50^\circ = 0.7660$, $\sin 55^\circ = 0.8192$, $\sin 60^\circ = 0.8660$, find $\sin 52^\circ$, using Newton's forward formula.

8. Estimate the value of $f(22)$ and $f(42)$ from the following data:

x	20	25	30	35	40	45
f(x)	354	332	291	260	231	204

Interpolation with unequal intervals:

Lagrange's interpolation formula:

1. Given the values

x:	5	7	11	13	17
f(x):	150	392	1492	2366	5202

Evaluate $f(9)$, using Lagrange's formula.

2. Find the polynomial $f(x)$ by using Lagrange's formula and hence find $f(3)$ for

x:	0	1	2	5
f(x)	2	3	12	147
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3. Use Lagrange's interpolation formula to find the value of y when $x = 10$, if the following values of x and y are given:

x:	5	6	9	11
y:	12	13	14	16

4. Given $\log_{10} 654 = 2.8156$, $\log_{10} 658 = 2.8182$, $\log_{10} 659 = 2.8189$, $\log_{10} 661 = 2.8202$, find the value of $\log_{10} 656$ by using Lagrange's formula,.

5. Using Lagrange's interpolation, calculate the profit in the year 2000 from the following data

Year:	1997	1999	2001	2002
Profit in Lakhs of Rs.	43	65	159	248

Numerical Differentiation:

1. Compute $f^1(x)$, and $f^{11}(x)$ at a) $x = 16$, b) $x = 15$, c) $x = 24$, d) $x = 25$ from the following table

x	15	17	19	21	23	25
$f(x) = \sqrt{x}$	3.873	4.123	4.359	4.583	4.796	5

2. Determine $y^1(2)$, $y^{11}(2)$ from the following data.

x	0	1	2	3
y	0	1	8	27

3. Compute y^1 , y^{11} at $x = 1.1$ and 1.5 from following data:

x	1.0	1.1	1.2	1.3	1.4	1.5
y	7.989	8.403	8.781	9.129	9.451	9.750

Numerical Integration:

1. Evaluate $\int_0^1 \frac{dx}{1+x^2}$ and $\int_0^6 \frac{dx}{1+x^2}$ by using

- (i) Simpson's one third rule
- (ii) Simpson's three eight rule
- (iii) Weddle's rule

By taking $n = 6$.

2. By using the Simpson's 1/3rd rule Evaluate: $\int_1^5 f(x) dx$

x	1	2	3	4	5
$f(x)$	13	50	70	80	100

3. Find the approximate value of $\int_0^{\pi/2} \sqrt{\cos \theta} d\theta$ by Simpson's 1/3rd rule by dividing $[0, \pi/2]$ into 6 equal parts.

4. Evaluate: $\int_0^1 \frac{x dx}{1+x^2}$ by Weddle's rule taking seven ordinates and hence find the value of $\log_e 2$.

5. Evaluate: $\int_4^{5.2} \log x dx$ by taking six equal parts using Simpson's 1/3rd, Simpson's 3/8th and Weddle's rule.

6. Using Simpson's 3/8th Evaluate: $\int_0^{0.6} e^x dx$ by taking 7 ordinates.